

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.2: Reflection — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.2: Reflection
Driving Question	DRIVING QUESTION: Why does a mirror show a perfect copy of the world — and where exactly does that copy live?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Symmetry Safari: Lines of Symmetry in Plane Figures	Students examined a range of plane figures — including cut-out shapes, Kenyan flag motifs, and leaf specimens — and physically folded or used a mirror strip to test for symmetry. They observed that some shapes (e.g., a rectangle, an equilateral triangle) could be folded so that both halves aligned perfectly, while others (e.g., a scalene triangle) could not. They recorded the number of lines of symmetry found for each figure.	A line of symmetry (mirror line) divides a plane figure into two congruent halves that are exact mirror images of each other. Regular polygons have a number of lines of symmetry equal to the number of their sides (e.g., an equilateral triangle has 3, a square has 4). A circle has infinitely many lines of symmetry. Figures with no such line are asymmetric. Pedagogical note: Ensure students distinguish between a line of symmetry and any line that simply cuts a shape in two — the halves must map onto each other perfectly.	The mirror image seen in a looking-glass is a real-world example of a line of symmetry: the mirror surface acts as the mirror line, and the reflected image is the congruent 'other half.' This anchors the driving question — the 'perfect copy' the mirror produces exists because reflection preserves shape and size exactly, just as a fold along a line of symmetry produces two identical halves. Teacher tip: Cold-call at least 3 students to state one line of symmetry for a shape on the board and justify their answer before moving on.
Lesson 2 Mirror, Mirror on the Wall — Properties of Reflection on a Plane Surface	Students placed a mirror strip vertically on dotted/grid paper and marked an object point P on one side. They measured the perpendicular distance from P to the mirror line, then located and measured the distance of the image point P' on the other side. They repeated this for multiple points at varying distances and angles. They also compared the orientation of an object letter (e.g., 'R') with its reflected image, noting the lateral flip.	Three formal properties of reflection on a plane surface were formalised: (1) The image distance from the mirror line equals the object distance from the mirror line (equidistance property). (2) The line joining an object point to its image is perpendicular to the mirror line. (3) Reflection preserves size and shape (isometry) but reverses orientation — the image is laterally inverted. Pedagogical note: Insist students state all three properties in their own words before recording the formal version; this surfaces misconceptions early.	These three properties explain precisely 'where the copy lives': the image of every point exists on the opposite side of the mirror, at an equal perpendicular distance. The mirror does not create something new — it maps each point to a unique, predictable location. Revisiting the driving question: the 'perfect copy' is perfect because all distances and angles are preserved; it lives at the perpendicular-equidistant position across the mirror line. Teacher tip: Ask students why a word on a matatu appears reversed in a rear-view

			mirror — wait 30 seconds before cold-calling 2 students.
Lesson 3 Drawing Images on a Plane Surface — Construction Practice	Students used a ruler and set square (or compass) to construct the reflection of given points, line segments, and polygons across a given mirror line m drawn on plain paper. They measured perpendicular distances from each vertex to m , marked the equidistant image point on the other side, and joined image points to form the image figure. They checked their constructions by folding the paper along m to verify overlap.	The image of any point P under reflection in mirror line m lies on the perpendicular from P to m , at a distance from m equal to the distance of P from m , on the opposite side. To reflect a polygon, students reflect each vertex individually and then join the image vertices in order. The construction process makes the equidistance and perpendicularity properties concrete and procedural. Pedagogical note: Circulate and check that students are drawing true perpendiculars — a common error is estimating rather than constructing the right angle.	Accurate construction practice reinforces that the mirror image's location is mathematically determined, not approximate. Every point in the 'copy' seen in a mirror has a precise geometric address. This lesson bridges the conceptual properties from Lesson 2 to the procedural skill of locating images, directly answering 'where exactly does that copy live?' with a ruler-and-set-square precision. Teacher tip: Have students annotate one completed construction with arrows labelling 'object distance' and 'image distance' to reinforce vocabulary.
Lesson 4 Reflection on the Cartesian Plane I — The x-axis and y-axis as Mirror Lines	Students plotted a triangle with given coordinates on the Cartesian plane and reflected it first in the x-axis, then in the y-axis. They recorded the coordinates of the object vertices and the corresponding image vertices in a table and looked for a numerical pattern in how the coordinates changed under each reflection.	Reflection in the x-axis maps every point (x, y) to $(x, -y)$ — the x-coordinate is unchanged and the y-coordinate changes sign. Reflection in the y-axis maps every point (x, y) to $(-x, y)$ — the y-coordinate is unchanged and the x-coordinate changes sign. These coordinate rules are a direct algebraic expression of the equidistance and perpendicularity properties established in Lesson 2. Pedagogical note: Use a two-column table (object image) prominently on the board so students can see the pattern emerge row by row before stating the rule.	Coordinate rules make 'where the copy lives' computable without drawing. A student can now state the exact address of any image point algebraically. This is the power of placing the mirror question on the Cartesian plane — it transforms a geometric idea into an algebraic rule. Teacher tip: Pose this check question and wait 20 seconds: 'If the point $(-3, 5)$ is reflected in the x-axis, what are the image coordinates?' Cold-call 3 students and ask each to justify the sign change.
Lesson 5 Reflection on the Cartesian Plane II — The $y = x$ Mirror Line	Students plotted the line $y = x$ on the Cartesian plane and reflected a set of given points across it. They recorded object and image coordinates in a table and identified the pattern: for every object point (x, y) , the image is (y, x) .	Reflection in the line $y = x$ maps every point (x, y) to the image point (y, x) — the two coordinates swap positions. The line $y = x$ acts as the mirror line; every object-image segment is perpendicular to $y = x$ and bisected by it. Students should verify this by checking that the midpoint of PP' lies on $y = x$ and that the gradient of PP' is -1 (perpendicular to the gradient of 1 for $y = x$). Pedagogical note: Many students initially guess the rule incorrectly; use the table of values to let the pattern emerge inductively before stating the rule deductively.	The coordinate-swap rule for $y = x$ is elegant and memorable, and it deepens the answer to the driving question: depending on which mirror (which line) is used, the algebraic address of the copy changes in a predictable, rule-governed way. The copy always lives at a location determined by the geometry of the mirror line — on the Cartesian plane, that geometry becomes arithmetic. Teacher tip: Ask students to describe in one sentence what 'swapping coordinates' means physically — cold-call 2 students after a 30-second think time.
Lesson 6	Students were given pairs of congruent	The mirror line is the perpendicular bisector	Finding the mirror line reverses the earlier

Finding the Mirror Line: Reverse Reflection on Plane Surfaces and the Cartesian Plane	figures (or object and image points) and asked to work backwards to find the mirror line. On plain paper they joined corresponding object and image points, constructed the perpendicular bisectors of those segments, and identified the mirror line. On the Cartesian plane they used midpoint and gradient calculations to determine the equation of the mirror line.	of the segment joining any object point to its corresponding image point. On the Cartesian plane: (1) the midpoint of PP' lies on the mirror line; (2) the gradient of the mirror line is the negative reciprocal of the gradient of PP'. Students can therefore find the equation of the mirror line algebraically using these two facts. Pedagogical note: Emphasise that any object-image pair gives the same mirror line — this is a useful self-check students should perform with at least two pairs.	process and deepens understanding: if you know where the 'copy' is, you can deduce exactly where the mirror must be. This completes the driving question from both directions — given the mirror, find the copy; given the copy, find the mirror. It also has a practical resonance: forensic scientists and engineers use analogous 'reverse reflection' reasoning. Teacher tip: Connect to a local context — ask students how a Nairobi traffic engineer might use this idea to position road mirrors at blind corners.
Lesson 7 Reflection and Congruence — Isometry and Invariant Properties	Students reflected various polygons across different mirror lines and systematically measured and compared: side lengths, interior angles, perimeter, and area of object and image figures. They also compared orientation (clockwise vs. anti-clockwise labelling of vertices).	Reflection is an isometry — it preserves distance (side lengths), angle measures, perimeter, and area. However, reflection reverses orientation: if the object vertices are labelled clockwise, the image vertices are labelled anti-clockwise, and vice versa. Properties that are unchanged by a transformation are called invariant properties. Points that lie on the mirror line are their own images (they are invariant points). Pedagogical note: The concept of invariant points is frequently tested; ensure students can identify that any point on the mirror line maps to itself.	The fact that reflection is an isometry is the precise mathematical reason why a mirror shows a 'perfect copy' — no stretching, no shrinking, no angle distortion. The driving question is now fully answered at a formal level: the copy is perfect because reflection preserves all metric properties; it is a copy (not the original) because orientation is reversed. Teacher tip: Ask students to identify one everyday situation in Kenya where a reversed image causes a practical issue (e.g., text in a rear-view mirror, a stamp impression) — take 3 responses after 30 seconds.
Lesson 8 Reflection in Real Life and Assessment Consolidation	Students examined real-world applications of reflection: bilateral symmetry in animals and Kenyan traditional patterns (e.g., Maasai beadwork, Kente-inspired designs), reflective road signage, and architectural symmetry (e.g., facades of Nairobi CBD buildings). They also completed a consolidation task requiring them to perform and describe reflections, find mirror lines, state properties, and write coordinate rules.	Reflection underpins symmetry in nature, art, architecture, and engineering. All the sub-strand's key ideas connect: lines of symmetry (Lesson 1), the three properties (Lesson 2), construction technique (Lesson 3), coordinate rules for x-axis, y-axis, and $y = x$ (Lessons 4–5), finding the mirror line (Lesson 6), and isometry/invariance (Lesson 7). Students should be able to select the appropriate tool (construction, coordinate rule, or perpendicular bisector method) depending on the context. Pedagogical note: Use the consolidation task formatively — identify students who confuse coordinate rules (e.g., swapping vs. negating) and re-teach with a targeted example.	The driving question — 'Why does a mirror show a perfect copy of the world, and where exactly does that copy live?' — is now fully answered: a mirror shows a perfect copy because reflection is an isometry (all distances and angles preserved); the copy lives at the perpendicular-equidistant position across the mirror line, whose exact coordinates can be determined by geometric construction or algebraic rule. Teacher tip: In the final 10 minutes, cold-call 4 students to each contribute one sentence to a whole-class spoken answer to the driving question, building a complete response collaboratively.